Auditable appendices

Read with report.md, the claim-level evidence ledger, and the complete candidate table. Evidence IDs identify a claim, not an independent publication count.

A. Case normalization and phenotype audit

Rhabdomyosarcoma

Verified HPO label. HP:0002859: Rhabdomyosarcoma

Interpretation retained. Reported diagnosis; subtype, molecular classification, active/remission status and therapy not known

Congenital nephrocalcinosis

Verified HPO label. HP:0000121: Nephrocalcinosis

Interpretation retained. ID captures calcium deposition, not congenital onset or cause; onset requires documentation

Short stature

Verified HPO label. HP:0004322: Short stature

Interpretation retained. Requires longitudinal age/sex-adjusted anthropometry; not synonymous with GH deficiency

Failure to thrive

Verified HPO label. HP:0001508: Failure to thrive

Interpretation retained. Reported poor growth; feeding, absorption, metabolic demand and cancer/treatment effects unresolved

Reduced muscle bulk/reported atrophy

Verified HPO label. HP:0003202: Skeletal muscle atrophy

Interpretation retained. Retain atrophy only if clinically established. HP:0003199: Decreased muscle mass is the more literal term for low bulk alone

Birth around 32 weeks

Verified HPO label. HP:0001622: Premature birth

Interpretation retained. Fits reported gestational age; exact obstetric dating unknown

Birth weight around 1 kg, reported SGA

Verified HPO label. HP:0001518: Small for gestational age

Interpretation retained. Approximate weight fits very-low-birth-weight terminology, but does not independently establish SGA or fetal growth restriction. Need sex, exact gestational age, weight and reference chart. Approximation near 1 kg cannot determine a strict below-1-kg category

Parental recurrent miscarriages

Verified HPO label. HP:0200067: Recurrent spontaneous abortion

Interpretation retained. Family-history annotation attached to the affected parent/pregnancies, not a phenotype of the child; no inferred carrier CIN mechanism

The exact downloaded ontology terms and definitions are in phenotype_terms.csv; snapshot provenance is retained in `reference/hpo_selected.json` [E25]. Distinguish prenatal/neonatal findings from growth or muscle changes arising during cancer, glucocorticoid exposure, inactivity, malnutrition, surgery, radiation or chemotherapy. No chronology beyond the supplied descriptions was inferred. Microcephaly was **not** supplied and was not added to the phenotype list.

Smallest decision-critical records to request through the clinical team

- Original laboratory report with assembly, transcript version, genomic/cDNA HGVS, genotype calls, assay scope and copy-number findings; laboratory-supported phase/parental segregation. No public upload is needed.
- Baseline cytogenetic evidence, tissue/source and mosaicism distribution; other relevant molecular diagnoses or unresolved variants. Low-level mosaicism can differ by tissue and assay.
- Cancer pathology and molecular classification, including FOXO1 fusion status where appropriate; tumor sequencing/fusion results, current disease status, prior/current treatment and toxicity chronology.
- Renal imaging timing plus serum creatinine/cystatin C as clinically appropriate, electrolytes, bicarbonate, calcium/phosphate/magnesium, urine pH, age-adjusted calcium/citrate/oxalate indices and medication history. Nephrocalcinosis does not establish kidney failure or normal clearance.

B. Extended variant mechanism and genotype branches

Public mapping versus case identity

The script verifies the public ClinVar nucleotide substitutions against RefSeq sequence and independently reconciles their chromosome positions through Ensembl exon intervals. It does not use a protein label to guess a DNA allele. Transcript length agreement is checked; the original laboratory transcript still takes precedence for case reconciliation. `data/reference_context.json` records reference codon, flanking sequence, exon number and explicit patient-identity uncertainty.

ClinVar VCV000533901 contains multiple submitted pathogenic/likely pathogenic assertions; VCV004600147 contains a single-submitter VUS assertion. Underlying clinical segregation details are insufficient to phase this case. The public stop record exposes several population-database annotations in `variation_set`; these are preserved verbatim in `data/population_annotations.json`, without combining releases into an invented frequency. An empty annotation list for another record does not mean absent from gnomAD. The current gnomAD browser/API allele record was not independently established in this search. [E01–E03]

p.Leu737Ter

- **NMD:** the public matching stop lies before multiple downstream junctions. This supports predicted NMD under the usual exon-junction model, not measured RNA destruction. Assess allele-specific RNA in native transcripts with and without a research NMD perturbation, with a known NMD-sensitive control. Tissue, translation and stress state can alter the effect.
- **Retained sequence:** N-terminal interaction regions and the KARD region precede the stop; the C-terminal annotated kinase-like domain is lost. Kinase-domain annotation is not evidence that an activating kinase drug would repair BUBR1. Binding, scaffolding, localization and checkpoint timing must be assayed separately. [E04, E07, E12, E20]

- **Residual product:** an expressed truncated protein could supply useful scaffolding, sequester partners, mislocalize, or be unstable. The BUBR1(1–730) construct result motivates a test but cannot establish any of these outcomes for an endogenous 1–736 product. [E12]
- **Readthrough:** verify native stop and flanks, transcript abundance, product size and the inserted peptide. Near-cognate insertion need not recreate leucine. Require rescue of localization/interactions and completed-division fidelity. Native/minigene assays are needed because intronless cDNA expression removes NMD context. [E18, E46]

p.Asn1002Lys

No retrieved exact-variant experiment established reduced abundance, destabilization, altered folding, binding, localization, checkpoint signaling, attachment control, a splice defect, or preserved function. These remain competing possibilities. Domain membership provides a hypothesis, not a prediction of pathogenicity or a drug target. Do not transfer effects of neighboring residues or a generic missense score to this allele. An exonic nucleotide change can affect splicing independently of its annotated amino acid; resolve the actual nucleotide and test RNA before considering splice correction. No splice-correcting approved medication was found for a demonstrated defect here. [E02–E04, E07]

Phase and additional findings

- **In trans:** one allele may undergo NMD while the other supplies most full-length protein; residual-function testing is decisive.
- **In cis:** the upstream stop may prevent routine translation of the downstream missense site on that same allele. Readthrough could then produce a full-length protein that still carries Asn1002Lys. The second homolog may be WT, carry a missed allele/CNV, or have altered expression. Revisit the recessive model rather than assuming compound heterozygosity.
- **Unknown phase:** retain both branches; do not presume a WT allele is available for upregulation.
- **Additional diagnosis/structural variant:** renal or muscle features may have another cause; laboratory re-review should cover the assay's blind spots without forcing a single-gene explanation.
- **Tissue-specific expression or mosaicism:** blood findings and cultured fibroblasts may not predict renal, muscle, progenitor or tumor states. Record passage, culture selection and karyotype drift.

C. Mechanism landscape: axes A to I

Axis A: Allele-specific restoration

Hypothesis (strategy 1): restoring native stop-allele translation produces useful BUBR1. **Required state:** correct stop context, sufficient RNA, functional readthrough product. **Anchors:** context-sensitive gentamicin reporters; macrolide/APC experiments; public stop sequence. **Approved entry points:** azithromycin, amikacin and gentamicin Tier3. Azithromycin is attractive because macrolides read through premature stops at roughly 100-fold lower concentration than gentamicin [E46]; amikacin may carry comparatively lower nephrotoxicity but, like all aminoglycosides, requires `MT-RNR1` screening (m.1555A>G, m.1494C>T, m.1095T>C) before use because carriers face a high risk of aminoglycoside-induced hearing loss [E53]. **Missing links:** exact-allele readthrough, inserted residue, correctly localized protein, fidelity, safe exposure. **Counterarguments:** NMD limits substrate; reporter signal can be artifactual; longer protein may fail; antibiotic concentrations at ribosomes are not whole-tissue accumulation; chronic renal/auditory/mitochondrial or QT/hepatic effects matter. **Test:** native transcript with orthogonal peptide and functional assays. **Stop:** no functional rescue before toxicity/exposure limits. Broad NMD inhibition is a laboratory probe only; global RNA-surveillance disruption and harmful truncated products prevent a systemic rationale. Splicing correction remains closed until an actual splice defect is demonstrated. [E01–E03, E18, E28, E30, E46, E50, E53]

Axis B: Preserve or increase residual functional BUBR1

Hypothesis (strategy 2): a useful but unstable allele can be stabilized, or its degradation slowed, to raise *functional* BUBR1. **Required state:** equal-expression complementation shows useful function; native protein half-life/localization are limiting. **Anchors:** allele-specific MVA defects; SIRT2-K668 stability; HSP-directed NPC experiments as remote context; autophagy-dependent BUBR1 turnover. **Approved entry points:** arimoclomol, hydroxychloroquine, metformin, trametinib and prescription niacin ER Tier3; broad proteasome inhibition (bortezomib) TierX. **Missing links:** BUBR1-specific chaperone dependence or selective NAD/SIRT2 response and mutant fidelity. **Contradictions and cautions:** arimoclomol is a heat-shock-response *co-inducer* (HSF1 → HSP70/HSP90), not an HSP90 inhibitor, and it did not induce HSP70/correct storage in another NPC cell study; FDA states its clinical NPC mechanism is unknown. Metformin changed BUBR1 **bidirectionally**: raised in some cell types and lowered in others within the same study, alongside glucose-dependent genotoxic stress [E23]. Hydroxychloroquine raised BUBR1 only in senescent cells by blocking autophagic/lysosomal degradation (not synthesis), and the autophagy inducer everolimus lowered it [E54]. Trametinib's rationale inverts a finding that BubR1 *deficiency* enhances MEK/ERK, so it is explicitly theoretical [E39]. SIRT2-K250 participates in checkpoint silencing; more deacetylation is not uniformly beneficial. **Test:** native protein turnover, K668/K250 state, localization and fidelity with target-dependence controls. **Stop:** protein-only increase, wrong localization, partner sequestration or tumor protection. No approved selective BUBR1 stabilizer/SIRT2 activator was established. [E07, E09–E11, E23, E37, E39, E47–E49, E51–E52, E54]

Axis C: Specific BUBR1 functions

Hypothesis: correct the demonstrated interaction defect rather than total abundance. **Required state:** distinguish MCC formation/BUB3 interaction/CDC20-APC-C regulation from PP2A-B56 attachment control. **Anchors:** domain complementation, B56-binding mutants and kinetochore-tethered rescue. **Approved entry point (strategy 3):** fingolimod is carried at Tier3 as an explicitly speculative PP2A-reactivation probe. It activates PP2A in *bulk* by antagonising the SET/I2PP2A inhibitor; no approved agent produces selective kinetochore B56 recruitment or MCC reconstruction, and fingolimod's approved S1P activity and the broader PP2A literature do not establish spatial recruitment. **Test:** co-immunoprecipitation/proximity readouts, kinetochore occupancy, cyclin-B/securin dynamics, cold-stable fibers and unperturbed divisions. **Stop:** only global phosphatase activity or prolonged arrest; no improved fidelity. [E07, E12–E14, E20, E36]

Axis D: Functional compensation/bypass

Hypothesis: rebalance attachment kinases/phosphatases without restoring BUBR1. **Required state:** attachment-control defect with a tractable opposing pathway. **Anchors:** targeted B56 rescue; partial Aurora-B-inhibitor rescue; U0126/ERK inhibitor rescue in RAS/MAPK-hyperactive cells. **Approved entry:** trametinib only as a conditional drug-substitution experiment. **Missing links:** excessive MAPK in normal case cells, clinically relevant drug effect, faithful divisions, window over tumor. **Counterarguments:** Aurora B is necessary for error correction; PLK1 inhibition did not rescue in the cited study and produced spindle abnormalities. Prolonged mitosis is not correction. MEK inhibition may improve malignant-cell mitotic fitness. **Test:** compare genetic bypass, research inhibitor, approved candidate and rescue-resistant/orthogonal perturbations. **Stop:** alignment without faithful anaphase, arrest, cell loss, or equal normal-cell toxicity. [E12–E14, E19, E39]

Axis E: Downstream genome/cellular injury

Hypothesis: a demonstrated consequence of complex aneuploidy is modifiable even if segregation errors persist. **Anchors:** BubR1-low oxidative-challenge sensitivity; fly SAC-loss proteostasis/mitochondrial dysfunction and genetic rescues. **Required state:** show the injury in relevant BUB1B normal tissue, not merely ROS or damage in an unrelated cancer. **Approved entry:** sirolimus is now deprioritised to TierX (fly cell-preservation signal only; preliminary negative mouse report); no sufficiently supported approved mitochondrial/chaperone/ROS-scavenger equivalent was identified. **Missing links:** which injury mediates which tissue phenotype, and whether a drug reproduces genetic rescue without preserving tumor cells. **Counterarguments:** DNA-damage pathways were not the main explanation of stemness loss in the fly model; CIN is not HR deficiency. Metformin's high-dose low-glucose genotoxic signal is context-specific contrary evidence, not proof of harm at usual exposure. **Test:** cell-state-matched replication-fork, mitochondrial, redox and proteostasis measures followed by orthogonal causal perturbation; productive tissue output. **Stop:** marker reduction without functional benefit; therapy antagonism; no causal injury. PARP inhibitors are not prioritized from 'genomic instability' alone. [E23–E24, E43–E45]

Axis F: Senescence, inflammation and tissue dysfunction

Hypothesis: senescent cells or stress-associated signaling drive a specific tissue deficit. **Anchors:** INK-ATTAC clearance in BubR1H/H mice; tissue-dependent p21/p53 effects. **Approved entry:** sirolimus and dasatinib both TierX (deprioritised) in this context. **Missing links:** harmful senescent population, selective pharmacologic removal or useful SASP suppression, productive tissue regeneration. cGAS-STING activation was not demonstrated for the reported genotype in retrieved evidence; micronuclei alone do not establish it. **Counterarguments:** genetic cell deletion is not a senolytic drug; senescence can restrain tumors; anti-inflammatory intervention may suppress antitumor immunity; progenitor deletion and growth inhibition matter in childhood. The preliminary rapamycin-null statement is low-rigor contrary evidence, not a definitive trial. **Test:** multiple senescence markers and persistent state, secretion assays, functional tissue coculture and selective perturbation. If cGAS/STING activation is measured, use genetic epistasis before any drug hypothesis. **Stop:** protective senescence lost, immune harm, progenitor injury, or inflammation reduced without tissue improvement. [E15–E17, E31, E33, E43]

Axis G: Tissue-specific support

Renal hypothesis: appropriate mineral/acid-base correction can reduce new deposition independently of BUB1B. **Required facts:** nephrocalcinosis timing/distribution; prematurity exposures including loop diuretics; hypercalciuria, hypocitraturia, oxalate, urine pH, acid-base state, calcium/phosphate/PTH/vitamin-D context as indicated, and tubular/filtration function. **Entry:** potassium citrate ER is deprioritised to TierX pending a defined citrate-responsive biochemical diagnosis and a feasible pediatric formulation. Hypercalciuria-directed or inherited hyperoxaluria therapies require their own diagnoses and regulatory review; they were not elevated into the shortlist without those facts. **Test:** nephrology-directed biochemical classification, mineral-specific supersaturation and experimental renal/crystal response if needed. **Stop:** no biochemical match, unsafe potassium handling or unfavorable calcium-phosphate supersaturation. [E35]

Muscle/growth hypothesis: reduced muscle mass and poor growth may reflect nutrition, prematurity, endocrine abnormalities, disuse, tumor burden or therapy rather than primary atrophy. Document strength, motor function and pre-treatment trajectory; use CK/metabolic/endocrine investigations only when clinically indicated. No approved BUB1B-specific muscle/growth medication was found. Prescription niacin evidence in adult mitochondrial myopathy does not establish this diagnosis; somatropin is contraindicated with active malignancy. A muscle model becomes justified only after a demonstrable muscle-level defect. [E38, E49, E52]

Axis H: Tumor vulnerabilities

Hypothesis: exploit a measured tumor dependency while sparing constitutional normal tissues. **Required state:** pathology/subtype; FOXO1 classification as relevant; tumor sequence/fusion/CNV profile; germline/tumor allele state; therapy history. **Entry (strategy 4):** larotrectinib Tier2 for a true NTRK-fusion dependency and matching clinical context (not NTRK expression or amplification alone); vincristine Tier2 as the standard RMS cytotoxic backbone with MVA-aware dose modification [E22]; olaparib Tier3 only if a tumor homologous-recombination/repair dependency is demonstrated: chromosomal instability alone does not imply HRD, and a PARP indication does not follow from BUB1B loss [E24, E55]. **Other searches:** temsirolimus trial evidence, MAPK-related BUBR1 dependency and generic CIN vulnerabilities were reviewed. **Missing links:** no tumor molecular data and no normal-cell window. DepMap/PRISM/GDSC sensitivity, if later accessed, is only a discovery aid. **Counterarguments:** checkpoint inhibition, mitotic stress, proteasome injury and DNA-damage targeting may hit affected normal tissues at least as hard; aneuploidy does not imply HRD. **Test:** matched tumor/normal panels, independent target perturbation, exposure/time dependence and current-therapy antagonism. **Stop:** no dependency, equal normal toxicity or tumor rescue by a constitutional-cell intervention. [E19, E21–E24, E32, E40–E42, E55]

Axis I: Combinations/sequences

No clinical combination is proposed. A **research** sequence is permissible only after single mechanisms work:

1. NMD-limited substrate followed by readthrough: use a factorial design to establish whether substrate availability genuinely enables functional rescue, not merely a larger dysfunctional band. Broad NMD blockade may be harmful.
2. Readthrough followed by protein-folding support: only if the readthrough product is demonstrably unstable yet functional at matched abundance. Test antagonism and total proteotoxic burden.
3. Correct segregation first, then address persistent tissue injury: distinguish prevention of new aneuploidy from surviving old aneuploid cells. Never claim reversal of an established tissue phenotype from an upstream assay.

All remain below supported single-mechanism hypotheses. Separate tumor and normal models are mandatory before considering translational compatibility. No combination advancement without feasible individual exposures, a genuine enabling effect, and absence of treatment antagonism.

D. Tier rules and exposure assessment

The exact candidate table is candidates.csv. A single major mechanistic, exposure or tumor-safety failure can reject a candidate; publication count and graph degree do not compensate.

- **Tier1:** genotype-compatible functional rescue in an appropriate model, substantially complete causal chain, plausible exposure, no invalidating unresolved safety conflict. Empty.
- **Tier2:** bounded decisive gaps with a plausible translational path. Here the tumour-directed options qualify conditionally (larotrectinib on a confirmed NTRK fusion, vincristine as the standard RMS backbone), contingent on tumour molecular findings and clinical context, not constitutional MVA rescue.
- **Tier3:** multi-link transfer, absent exact-genotype function, uncertain exposure or exploratory tissue mechanism.
- **TierX:** wrong direction, no meaningful bridge, major safety/window problem, or contradicted generalized benefit claim. Reopening criteria remain recorded.

Evidence dimensions are independent: **rigor** (design/controls), **causal directness** (perturbation and rescue versus association), **variant/model relevance**, **exposure feasibility**, **reproducibility** (independent versus same family), **clinical-context relevance**. High/moderate/low/unknown/not-applicable refer to each dimension; they are not averaged. Abstract-only access limits rigor assessment. Regulatory records are strong sources for approved use and warnings, not molecular rescue experiments.

Exposure bridge not completed for any constitutional candidate. Compare eventual effective *unbound intracellular* concentrations and duration with human pharmacokinetics, tissue distribution, protein binding,

metabolite activity and pediatric clearance. Nominal dish concentration, fly-food concentration and total tissue accumulation cannot be substituted for that bridge. Published values retained for context include gentamicin 800 micrograms/mL in reporters, NMN 1 mM for 24 h in MEFs, Aurora tool ZM447439 2 micromolar for 1 h under arrest, and rapamycin 40 micromolar in fly food for about 48 h. These are source-reported experimental conditions, exported by `scripts/build_package.py`; none is converted to a child's dose. [E10, E13, E18, E43]

E. Graph interpretation and missing-link audit

paths.csv lists exact edge sequences, conservative unproven-link counts, inferred-link counts, contradicted links, transfer events and decisive experiments. Species/tissue/genotype transfers are deduplicated labels within a path, not distances or probabilities. Any non-measured edge is counted as **unproven for a complete intervention chain**; an evidence-backed inference is therefore transparently stricter than an experiment. Model-measured edges still carry case-transfer requirements.

Exact Leu737Ter or Asn1002Lys functional rescue

Search outcome. Exact aliases/nucleotide records and primary BUB1B studies searched; no matching experiment retrieved; supplements incompletely inspected

Consequence. Retain uncertainty, not 'no function'

Approved SIRT2/NAD intervention → useful mutant BUBR1

Search outcome. SIRT2/NAD/stability and niacin bridge searched; abundance anchors found, full functional chain absent

Consequence. Tier3; no selective-activation claim

Arimoclomol → BUBR1 folding/function

Search outcome. Direct bridge search returned no candidate publication; remote NPC studies include contradiction

Consequence. Tier3; native allele test decisive

Fingolimod → kinetochore B56 recruitment

Search outcome. Targeted PP2A/BUBR1 bridge search found no localized rescue

Consequence. Tier3 strategy 3, explicitly speculative; bulk PP2A is insufficient for localized recruitment

MEK suppression → BUBR1-deficient fidelity

Search outcome. Relevant U0126 alignment/kinetochore study found; not approved-drug/case transfer

Consequence. Conditional Tier3

Rapamycin → functional tissue rescue

Search outcome. 2026 partial cell-count evidence found; related genetic tissue endpoint negative; older preliminary mouse-negative statement

Consequence. Preserve partial evidence, do not promote to tissue benefit

cGAS-STING → case tissue injury

Search outcome. Search did not retrieve genotype-specific causal evidence

Consequence. Hypothesized edges only; not a drug shortlist anchor

Readthrough → functional native BUBR1 at safe exposure

Search outcome. Context and nonBUB1B drug evidence found; exact bridge absent

Consequence. Strategy 1: azithromycin, amikacin and gentamicin Tier3

Signs are conditional, not simply multiplied. Aurora/PP2A balance is nonlinear; checkpoint inhibition can permit errors while excessive checkpoint persistence causes arrest. SIRT2 effects depend on residue and timing. Tumor and normal-cell endpoints never share an unlabeled context. A long path was retained for niacin because the approved-precursor entry adds a distinct question to the NMN evidence, not because extra hops improve credibility.

F. Conditional decision matrix

1

Condition. Stop undergoes NMD; missense has useful residual function

Favored investigation. Preserve missense only if abundance/stability limiting

Disfavored mechanism. Increasing degraded stop output without a functional product

Required test / ranking consequence. Native RNA, protein half-life and matched-expression complementation; stabilization may move up

2

Condition. Missense abundant but functionally defective

Favored investigation. Specific interaction/localization rescue or bypass

Disfavored mechanism. Abundance-only strategy

Required test / ranking consequence. MCC/B56/localization and fidelity; downgrade arimoclomol/niacin unless conformational rescue demonstrated

3

Condition. Truncation escapes NMD and has partial function

Favored investigation. Titrated residual-scaffold assessment

Disfavored mechanism. Automatic NMD inhibition

Required test / ranking consequence. N/C antibodies, co-IP and dosage-response; stop if interference worsens fidelity

4

Condition. Neither allele is sufficiently rescuable

Favored investigation. Demonstrated bypass or tissue-specific downstream mechanism

Disfavored mechanism. Generic BUBR1 upregulation

Required test / ranking consequence. WT correction versus each allele; preserve no-drug conclusion if bypass also fails

5

Condition. In cis or phase unresolved

Favored investigation. Revisit molecular model, other allele/CNV and diagnosis

Disfavored mechanism. Unconditional allele-specific priority

Required test / ranking consequence. Segregation/phasing; readthrough in cis may retain missense change

6

Condition. Readthrough yields protein but no functional rescue

Favored investigation. Investigate inserted residue/folding only if actionable

Disfavored mechanism. Continued readthrough escalation

Required test / ranking consequence. Peptide sequence, localization and fidelity; reject protein-only result

7

Condition. Checkpoint marker improves but arrest/death/senescence rises

Favored investigation. Identify nonproductive mitotic delay

Disfavored mechanism. Calling checkpoint timing rescue

Required test / ranking consequence. Completed-division rate, survival, persistent senescence; fail advancement

8

Condition. Micronuclei fall only through slower division or cell loss

Favored investigation. Analyze lineage selection explicitly

Disfavored mechanism. Claiming correction

Required test / ranking consequence. Per-division live-cell errors, death timing and karyotype distributions; no promotion

9

Condition. Normal-cell markers improve but tumor grows faster

Favored investigation. Context-specific hazard characterization

Disfavored mechanism. Systemic rescue hypothesis

Required test / ranking consequence. Paired tumor proliferation/clonogenicity and therapy response; major stop signal

10

Condition. Tumor drug equally toxic to affected normal cells

Favored investigation. Seek different dependency/route

Disfavored mechanism. Assumed therapeutic window

Required test / ranking consequence. Exposure-matched tumor/normal curves; reject window claim

11

Condition. Nephrocalcinosis with unknown filtration/tubular function

Favored investigation. Biochemical diagnosis and renal evaluation

Disfavored mechanism. Assuming either renal failure or normal clearance

Required test / ranking consequence. Electrolytes, acid-base/urine chemistry and filtration assessment; citrate/gentamicin safety unresolved

12

Condition. Receiving anticancer therapy

Favored investigation. Regimen-specific interaction and toxicity research

Disfavored mechanism. Independent systemic-drug interpretation

Required test / ranking consequence. Marrow/renal/immune/hepatic/mitochondrial overlap, CYP/transporter/QT effects and antagonism tests

13

Condition. Relevant exposure exceeds established human safety constraints

Favored investigation. Mechanistic probe only or alternative node

Disfavored mechanism. Advancing an elegant but infeasible drug path

Required test / ranking consequence. Unbound PK/PD and toxicity comparison; downgrade regardless of target engagement

G. Evidence, rejection, patent and dataset ledgers

The separate research-tool list keeps genetic perturbations, unapproved compounds and historical regulatory exclusions outside the approved-medication shortlist. The load-bearing claim audit records supporting passages and prevents model-specific measurements from becoming full-chain efficacy claims. For the renal branch, the Urocit-K label explicitly notes nonrandomized, non-placebo-controlled pivotal studies and potential dietary confounding; this supports cautious biochemical hypotheses rather than a congenital-nephrocalcinosis efficacy claim. [E35]

`data/evidence.csv` includes access category and read scope. **Full-text sections** means the cited results/methods/captions were read, not every figure image, supplement or raw dataset. Some DOI downloads served metadata, a preprint or invalid PDF content; these were not silently treated as primary full text. The published NPC arimoclomol-negative PMC text was read after an initial preprint copy was encountered. The 2026 fly PDF was an accepted/article-in-press version under the published DOI. Quantitative interpretations stay within inspected results.

Corrections and study families: the JCI allele study and its correction are one study family. ARST1431 publication and trial registry are one study, not replication; registry status on retrieval was active-not-recruiting, reflecting continued follow-up. SIRT2/NMN/genetic-lifespan observations share one paper/program. PP2A studies may reuse established MVA lines from earlier cohorts; independence for the exact genotype is not claimed. Macrolide and niacin corrections were identified but their substantive content was not fully available; no precise drug effect magnitude from those corrected studies drives a tier.

Patents: Google Patents full-text targeted searches inspected WO2017062311A1 (CD38/NAD-related family) and WO2017049409A1 (readthrough family). The former cites the BUBR1/SIRT2 work rather than supplying independent BUB1B rescue; the latter yielded no BUB1B working example in the targeted text inspection. National family members are not counted as replication. Espacenet, WIPO PATENTSCOPE and Lens were not independently searched; no completeness, legal-status or freedom-to-operate claim is made. [E27]

Datasets: GSE67905 is a HeLa siRNA/mitotic-slippage expression dataset, not a drug-rescue screen. The 2026 fly article reports GSE290206 (CIN) and GSE301548 (trisomies); accessions were read from the paper, but raw matrices and source-data tables were not reanalyzed. DepMap26Q3 release availability was verified; PRISM/GDSC/CTRP panels were not analyzed. Proposed analyses must track genotype, passage, cell cycle, proliferation, concentration, time, batch and biological replicates; signature reversal would remain insufficient for rescue. datasets.csv

`data/source_coverage.csv` separates accessed sources from unsearched/unverified resources. Public retrieval succeeded for ClinVar, RefSeq/Ensembl/MANE, UniProt, HPO, PubMed/Europe PMC/selected PMC/publisher texts, FDA/DailyMed/EMA, one trial registry, selected Google Patents and dataset metadata. NCBI Gene HTML and some GEO HTML requests returned access obstacles; reference/metadata alternatives were used. ChEMBL, BindingDB, IUPHAR, Reactome, Open Targets, DGIdb, DrugCentral, LINCS, cBioPortal, PRIDE and other listed resources were not comprehensively queried or licensed for bulk use. No claim of all-approved-drug or all-patent coverage is made.

Stopping rule: the bounded search closed after variant triage, forward mechanistic retrieval, backward approved-drug/label checks, targeted missing-link/counterevidence searches and the material 2026 bridge update. Remaining gaps predominantly require exact allele data, experiments, exposure studies or unavailable full texts, rather than another generic co-mention search. Formal database saturation was not established. The selected query log plus all cited-source retrievals are in `data/search_log.csv`; discovery-service internal subqueries are not available as a complete export.

H. Minimum discriminating experimental program

H1: Highest information gain: allele-resolved function

Materials: authorized patient-derived normal cells if available, independent isogenic corrected clones and parental controls where authorized; otherwise a clearly labeled engineered model of the *verified nucleotide alleles*. Do not engineer a guessed protein-equivalent nucleotide variant. Include WT/WT, each allele separately, the phased pair, BUB1B-deficiency control and corrected WT benchmark. Native genomic editing preserves NMD/splicing; complementary inducible cDNA expression tests protein function but cannot assess NMD.

Stage 1, native output: allele-specific RNA with transcript-junction assays and an NMD-sensitive positive control; NMD perturbation as a short research diagnostic only. N- and C-terminal protein detection plus targeted proteomics distinguish full-length and truncated species. Measure stability and kinetochore occupancy normalized to mitotic cell state. Test splice junctions rather than assuming the missense annotation exhausts the consequence.

Stage 2, function at matched expression: complement each allele at physiologically calibrated abundance; compare WT correction. Assay BUB3/CDC20/MAD2 complex formation and cyclin-B/securin destruction timing; separately B56 localization, K-fibers and attachment correction. Short drug-induced checkpoint challenges can diagnose function, but primary rescue measurement occurs in unperturbed completed divisions.

Primary endpoint: fraction of completed mitoses with prospectively defined lagging chromosome/bridge/missegregation events, assessed by blinded live-cell imaging. Report total entry into mitosis, completion, arrest, death, generation time and daughter-cell fate separately. Confirm a subset with single-cell chromosome measurements; micronuclei alone are secondary. Track lineages to distinguish fewer new errors from removal of existing abnormal cells.

Prospective proposed advancement criterion, not observed evidence: a candidate must close at least half of the mutant-versus-corrected excess error rate, with a confidence interval excluding no improvement, without a greater-than-10% decrease in completed divisions or viable daughter output. These are screening gates requiring pilot refinement before unblinding, not clinically meaningful benefit thresholds. If corrected cells do not establish a reproducible assay deficit, do not compute a rescue fraction or screen on noise.

Define the proposed rescue fraction as `(p_vehicle - p_candidate) / (p_vehicle - p_corrected)`, where each probability is the new-error rate per completed division in the matched biological run. Require a positive, reproducible denominator; report absolute risk difference alongside this fraction and do not truncate negative or greater-than-one results. Arrest and death remain separate outcomes, never removed to improve the numerator.

Reproducibility/statistics: at least three independent culture/imaging runs and two independent edited clones where feasible; biological run/clone is the unit, cells are nested observations. Pre-specify one primary contrast; use a binomial mixed model or cluster-bootstrap interval for per-division errors. Report raw effect, confidence interval and multiplicity-adjusted exploratory comparisons separately; use Holm adjustment for the limited predeclared mechanistic contrasts. Determine sample size from blinded pilot run/clone variability and baseline event rate rather than fabricating a power estimate. No p-value has been computed for proposed experiments.

Invalidation: WT correction fails to normalize the assumed defect; native allele identity differs; phenotype arises only from overexpression/culture selection; 'rescue' comes from arrest/death; or improved normal-cell fidelity accompanies tumor protection.

H2: Discriminate stabilization from attachment bypass

Rank next because it can eliminate whole drug classes cheaply. Compare expression-matched missense/truncation rescue with kinetochore-targeted B56, binding-defective B56/BUBR1 controls and untethered B56. If tethering improves alignment, require subsequent completed-division fidelity and sustained viable progeny. If protein increase works but tethering does not, prioritize native stability/localization assays. If tethering works despite nonfunctional allele output, bypass is favored. If both fail while WT correction works, another BUBR1 function is limiting.

Drug mapping: arimoclomol/niacin follow only an abundance/stability-limited branch; trametinib follows only a demonstrated MAPK-dependent branch. Compare target perturbation genetically and pharmacologically, including relevant inactive/binding-defective controls. Avoid concluding that generic proteasome inhibition or bulk PP2A activity reproduces a spatially specific rescue.

H3: Native stop readthrough

Use the verified allele with its introns/flanks; include WT sense construct, stop construct, translation/measurement controls and a native RNA abundance readout. Compare azithromycin to gentamicin as a mechanistic benchmark at analytically measured exposures, not a clinical regimen. Identify the readthrough peptide across the stop, quantify full-length product and test function under H1. A rise in C-terminal antibody signal alone is not sufficient. Test NMD rescue separately; a factorial experiment only follows evidence that substrate is limiting. Stop at a nonfunctional product or an implausible exposure window. Current corrections/full-method gaps in macrolide literature must be resolved before relying on its numerical pharmacology.

H4: Tissue-specific downstream discrimination

Only use a tissue model justified by a documented phenotype. In a demonstrated proteostasis/mitochondrial defect, compare genetic mitochondrial chaperone/ROS-scavenger support, appropriate autophagy perturbations and sirolimus. Measure flux rather than static LC3/lysosome accumulation, respiration/redox with cell-number controls, lineage output and actual tissue function. The fly result predefines the failure mode: **more surviving progenitors without productive tissue output is not tissue rescue**. In renal support, biochemical classification and mineral-specific supersaturation precede organoid pharmacology. Established deposits, new deposition and filtration are separate endpoints.

H5: Tumor and exposure gates

Current RMS tissue/organoid or a justified molecularly matched model must be compared with constitutionally affected normal cells, including a vulnerable tissue when indicated. Tumor subtype/fusion/CNV and target-dependence controls are required. Test effects on tumor growth, clonogenic recovery and the actual anticancer regimen once known; do not invent a drug combination. Model clinically plausible free-exposure/time windows, with renal, mitochondrial, marrow and immune relevance. Define failure as no selectivity, normal-progenitor loss, tumor protection, therapy antagonism or a required exposure outside established human safety constraints.

I. Up to three mechanism-distinct submission concepts

Concept 1: Native-allele rescue with a functional gate

Hypothesis: one allele supplies a rescuable product; its limiting step can be distinguished experimentally. **Fit:** directly addresses the stop/missense pair without presuming two trans alleles or a WT allele. **Evidence:** native context/NMD inference, variant-dependent BUBR1 dysfunction and nonBUB1B readthrough/chaperone anchors. **Candidate:** azithromycin only for a verified readthrough-compatible branch; arimoclomol only for a proven instability branch; both Tier3, not coadministration. **Gaps:** native product, phase, inserted residue, functional gain and exposure. **Validation:** H1 followed by H3 or the relevant stability test, not both by default. **Safety barrier:** tumor fitness plus antibiotic/chaperone systemic effects. **Novelty:** selection by allele-level failure mode, not BUB1B-expression signature. **Scalability:** reusable native-transcript/isogenic assay; each allele still needs validation. **Limit:** no exact-allele drug rescue presently demonstrated.

Concept 2: Localization-specific attachment bypass

Hypothesis: deficient local PP2A-B56 activity drives a demonstrable attachment defect even when total protein manipulation fails. **Fit:** putative truncation retains the recruitment region in sequence; MVA-model tethering offers a causal reference. **Evidence:** genetic B56 targeting, binding-defective controls and chromosome-alignment rescue. **Candidate:** no approved selective recruitment drug found; trametinib is a separate conditional bridge only with excessive MAPK. **Validation:** H2 plus completed-division fidelity and target epistasis. **Safety barrier:** excessive attachment stabilization or tumor mitotic protection. **Novelty:** spatial function rather than generic kinase or abundance modulation. **Scalability:** recruitment/fidelity screening can identify new compounds, but a screening platform is not an approved-drug answer. **Limit:** alignment and druggable localization remain decisive gaps.

Concept 3: Productive tissue rescue, not aneuploid-cell preservation

Hypothesis: a documented mitochondrial/proteostasis consequence is more tractable than the initiating segregation defect. **Fit:** independent of which allele carries residual function, but requires a relevant tissue lesion. **Evidence:** 2026 SAC-loss fly genetics distinguish mitochondrial support from TOR-associated cell preservation; BubR1-low oxidant sensitivity supplies a related, narrower anchor. **Candidate:** sirolimus as a deprioritised (TierX) comparator only; no justified approved equivalent of mitochondrial genetic rescue yet. **Validation:** H4, comparing productive progeny and tissue function rather than cell counts; maintain H5 tumor gate. **Safety barrier:** preserving damaged/premalignant cells, immunosuppression and developmental growth inhibition. **Novelty:** explicitly separates survival from useful tissue output and from chromosome correction. **Scalability:** flux/lineage assays across aneuploid disorders, with genotype/tissue revalidation. **Limit:** fly bub3/rod neural data do not establish BUB1B muscle/renal benefit.

No numerical hackathon score is assigned. If the rubric is later used, first exclude proposals failing mechanistic/safety gates; then anchor scientific rigor to controls and falsifiability, impact to the demonstrated unmet endpoint, innovation to a resolvable new mechanism, and scalability to reproducible assays. The 35/25/25/15 weights must not be interpreted as probability of treatment success.

J. Reproducibility, privacy and final quality check

Run `python scripts/normalize_references.py`, `python scripts/build_package.py`, then `python scripts/validate_package.py` from a fresh process. Standard library only; no patient files or network required for regeneration. Original extraction decisions reside in `curation/`; manual report interpretation is not generated by a language model at rerun. `data/search_log.csv` records selected discovery queries and cited-source retrievals, with source access and limitations. `data/validation.json` reports final schemas, reference integrity, counts and file hashes.

The GitHub layout is `docs/` (manuscript), `data/` (ledgers), `curation/` (JSON inputs), `reference/` (public snapshots) and `scripts/`. Optionally add future **de-identified, authorized** assay summaries and notebooks in separate versioned analysis folders. Keep controlled genomic data, pedigree identifiers, medical records, consent material and raw patient files outside the repository. Do not open-license third-party full texts or restricted datasets. No public push was performed; this package is intended for the research team and requires a publication/privacy review before any release because a rare phenotype combination can itself be identifying.

Unresolved questions: exact alleles/phase; useful residual function and dominant molecular defect; current tumor/treatment and renal physiology. **Next experiment:** H1. **Strongest surviving hypothesis:** a demonstrated scaffold/attachment deficit can be rescued experimentally, potentially by localized B56 bypass. **Main failure:** no useful residual product or apparent rescue without faithful, productive divisions and a tumor-safe window. **Conclusion-changing evidence:** reproducible exact-genotype functional/tissue rescue at feasible exposure with appropriate target-dependence and paired normal/tumor validation. Until then the full approved-drug causal chain remains incomplete.